

Confidence-Gated Multi-Agent Fall Detection: Retrieval-Augmented Reasoning and Cost-Sensitive Response on Wearable IMU Data

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Abstract—Wearable inertial fall detectors achieve strong average accuracy but remain brittle on ambiguous near-fall activities and provide neither calibrated uncertainty routing nor an explicit response policy under asymmetric false-negative cost. We present a *Risk-Calibrated Dual-Process* (RCDP) architecture—a confidence-gated multi-agent wrapper around a fixed Tier-1 detector (CNN-LSTM-Attention following Bhatti *et al.*)—with (i) a dual-threshold confidence gate, (ii) retrieval-augmented k -nearest-neighbor case memory, (iii) local large-language-model (LLM) chain-of-thought reasoning over biomechanical evidence, and (iv) a cost-sensitive action agent. On an *ambiguous stress benchmark* (SisFall D18/D19 near-falls, $n=300/\text{fold}$), Tier-1 F1 collapses from 0.713 to 0.896 with the full stack ($\Delta\text{F1} = +0.183$, $\Delta\text{Recall} = +0.208$, 84% cost reduction). Under subject-independent five-fold cross-validation on the main protocol ($n=1000/\text{fold}$), mean F1 improves from 0.759 ± 0.014 to 0.887 ± 0.029 and recall from 0.759 ± 0.022 to 0.977 ± 0.037 , with normalized response cost reduced by approximately 83%. External validation on binary KFall confirms cross-domain stability (F1 $0.987 \rightarrow 0.990$) rather than large headline gains. Paired ambiguous-case comparisons show Mistral substantially outperforms a heuristic reasoner (mean F1 0.972 vs. 0.034). We position results as gains over a fair reimplement of the Bhatti-style detector under our protocol, not as a claim against the original KFall multi-phase tables.

Index Terms—Fall detection, wearable IMU, confidence gating, retrieval-augmented reasoning, large language models, cost-sensitive decision making, SisFall, selective prediction.

I. INTRODUCTION

Population aging and independent living motivate continuous fall monitoring with wearable inertial measurement units (IMUs) [1]–[4]. Modern deep detectors can reach competitive accuracy on curated benchmarks [5], yet practical systems still fail in three ways that matter for safety-critical deployment. First, near-fall activities (e.g., sitting with compensatory motions) induce ambiguous windows that a softmax classifier force-labels [6]. Second, edge models rarely expose calibrated uncertainty that can defer difficult cases [7]–[9]. Third, a binary fall/ADL prediction is not a response policy: emergency escalation, caregiver notification, and logging have unequal costs, especially when false negatives dominate clinical risk [10].

Bhatti *et al.* [11] propose a strong hybrid CNN-LSTM-Attention detector for multi-phase fall analysis on wearable IMUs (“Beyond Falls”). Their contribution is primarily *detection*. We ask a complementary question: *given a fixed Tier-1 detector, can an agentic wrapper improve the safety operating point—raising recall and lowering expected response*

cost—while providing structured explanations on escalated windows?

This paper answers affirmatively with an RCDP pipeline: a fast *System 1* edge reflex (Tier-1 detector, ≈ 7 ms) routes confident windows deterministically, while ambiguous windows escalate to a deliberative *System 2* (evidence serialization, k -NN memory, local LLM CoT, cost-sensitive Action Agent). Uncertain predictions are escalated to an evidence serializer, a growing k -NN memory of past cases, and a local LLM reasoner that emits JSON rationales [12]–[14]; an Action Agent then maps decisions to simulated response tiers under asymmetric clinical cost $\mathcal{R}_\beta = \beta \cdot \text{FN} + \text{FP}$ with $\beta=10$. Our primary evaluation is SisFall [4] with subject-independent five-fold cross-validation, forced inclusion of near-fall codes D18/D19, and a homogeneous evaluation budget of 1000 stratified windows per fold. We additionally report an ambiguous stress benchmark ($n=300/\text{fold}$), larger-budget archives ($n=4000$), multi-seed training variability, latency profiles, and completed binary KFall external validation [15].

a) Contributions.:

- A complete agentic architecture—gate, evidence, retrieval, LLM CoT, action—wrapped around a Bhatti-style Tier-1 backbone without modifying its weights at inference.
- A locked paper protocol (splits, calibration, cost, near-fall metrics) that enables fair ablations: `tier1_only` $\rightarrow$ `gate_only` $\rightarrow$ `gate_knn` $\rightarrow$ `gate_llm` $\rightarrow$ `gate_knn_llm`.
- Five-fold SisFall evidence that the full stack improves F1/recall and normalized cost on every fold, with an **ambiguous stress benchmark** (D18/D19) where Tier-1 F1 is 0.713 vs. 0.896 for the stack ($\Delta\text{F1} = +0.183$).
- Paired LLM backend comparisons on shared ambiguous IDs, budget-sensitivity figures, binary KFall external validation, and latency characterization of the escalated path.
- An honest positioning relative to Bhatti *et al.*: gains are versus the *same* Tier-1 checkpoint under our SisFall protocol; we do not claim to supersede the original KFall multi-phase tables.

II. RELATED WORK

A. Wearable Fall Detection

Classical threshold detectors [16] and shallow learners have been succeeded by CNN, LSTM, TCN, and Transformer

variants on public corpora such as SisFall [4], KFall [15], UP-Fall [17], and UniMiB SHAR [18]. Surveys emphasize the trade-off between sensitivity and false alarms in free living [1], [3], [5]. Bhatti *et al.* [11] integrate CNN, LSTM, and attention for pre-/transition-/post-fall phases on KFall. We reuse their architectural motif as Tier-1 and add decision-time agents rather than proposing another detector backbone as the sole contribution.

B. Uncertainty, Calibration, and Selective Prediction

Modern networks are often miscalibrated [7], [19]. Selective classification and abstention [8], [9] motivate routing low-confidence samples to stronger (or human) decision makers. Our dual-threshold gate implements selective escalation tailored to fall cost asymmetry, rather than a single reject option.

C. Retrieval-Augmented and Agentic LLM Reasoning

Retrieval-augmented generation [13] and compact encoders such as MiniLM [20] support case-based reasoning. Chain-of-thought prompting [12] and reason-act loops [14] improve structured decisions. Local LLM runtimes [21]–[23] enable offline CoT without cloud APIs. We apply these ideas to IMU-derived biomechanical evidence rather than text-native tasks, and we close the loop with a cost-sensitive Action Agent.

III. METHOD

Fig. 1 summarizes the pipeline.

A. Tier-1 Detector

Given a window $\mathbf{x} \in \mathbb{R}^{C \times T}$ (here $C=6$, $T=90$ on Sis-Fall), a backbone f_θ produces logits and embedding $\mathbf{e}$. We convert logits to fall probability $p(\text{fall})$. The primary backbone is CNN–LSTM–Attention following Bhatti *et al.* [11]; we additionally train a zoo (CNN1D, LSTM, CNN–LSTM, TCN, ModernTCN [24], TSMixer [25], PatchTST [26], Transformer [27]) under the same protocol for context (Table XIII).

B. Confidence Gate

A dual-threshold gate with calibrated $(\tau_{\text{low}}, \tau_{\text{high}})$ routes windows:

$$\begin{cases} \text{ADL}, & p \leq \tau_{\text{low}}, \\ \text{Fall}, & p \geq \tau_{\text{high}}, \\ \text{Ambiguous}, & \text{otherwise.} \end{cases} \quad (1)$$

Thresholds are calibrated on subject-held-out training windows. Mode `tier1_only` disables meaningful escalation by collapsing the ambiguous band to a point decision at 0.5.

C. Evidence Serializer

For escalated windows we compute biomechanical descriptors (free-fall duration, impact magnitude, post-impact stillness, tilt change, peak gyroscope) and serialize them into natural-language evidence for retrieval and prompting.

TABLE I: System 2 escalation duty cycle by evaluation protocol (honest reporting). High rates on the ambiguous stress benchmark reflect deliberate near-fall stress; KFall external transfer invokes deliberation on $\approx 1\%$ of windows.

Protocol	n/fold	Escalation rate
SisFall main (5-fold mean)	1000	0.47 (± 0.08)
SisFall ambiguous stress	300	0.59
KFall external (5-fold mean)	1000	0.011

D. Retrieval Memory

A k -NN memory stores MiniLM [20] embeddings of evidence text with labels and features. At escalation time we retrieve $k=5$ neighbors (cosine similarity) as few-shot context, following the RAG principle [13].

E. LLM Reasoner

An LLM reasoner (`backend` $\in \{\text{ollama}, \text{heuristic}\}$) consumes evidence and neighbors and returns a JSON decision with rationale [12]. We evaluate heuristic, Mistral [22], and Qwen2.5 [23] on paired ambiguous IDs. Production full-stack rows use Ollama [21] with Mistral as the default local model.

F. Action Agent and Cost

The Action Agent maps severity cues to simulated actions (emergency / notify / monitor / log). Expected response cost uses

$$\text{Cost} = 10 \cdot \text{FN} + 1 \cdot \text{FP}. \quad (2)$$

We report *cost per 1000 windows* so folds with equal evaluation budgets are comparable. The FN weight encodes the clinical preference that missed falls are substantially more expensive than false alarms [10].

G. System Modes

Ablations instantiate cumulative agents: `tier1_only`, `gate_only`, `gate_knn`, `gate_llm`, and `gate_knn_llm` (full stack). The design isolates whether gains require retrieval, language-model reasoning, or both.

H. Deployment Workflow

Fig. 2 illustrates the RCDP dual-process deployment model. *System 1* executes on-device reflex classification (≈ 7 ms per window): confident ADL and fall decisions are logged or alarmed without cloud invocation. Ambiguous windows escalate to *System 2*, which operates during the post-impact immobility window (≈ 10 – 60 s) using biomechanical evidence $\Sigma(w)$, retrieval-augmented reasoning, and optional contrastive critique before emitting a graded action policy. Table I reports honest escalation duty cycles by protocol.

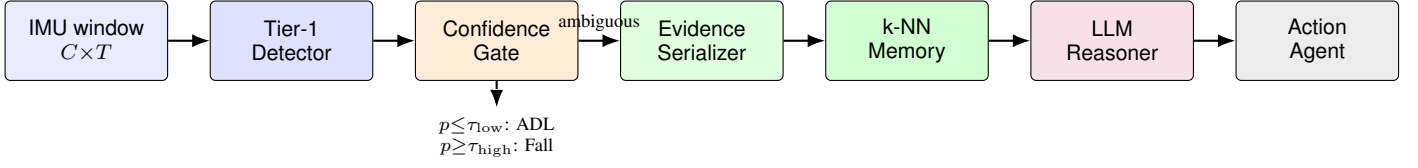


Fig. 1: Confidence-gated multi-agent fall detection. High/low confidence windows are decided by the Tier-1 detector; ambiguous windows escalate to evidence serialization, k -NN retrieval, LLM reasoning, and a cost-sensitive Action Agent.

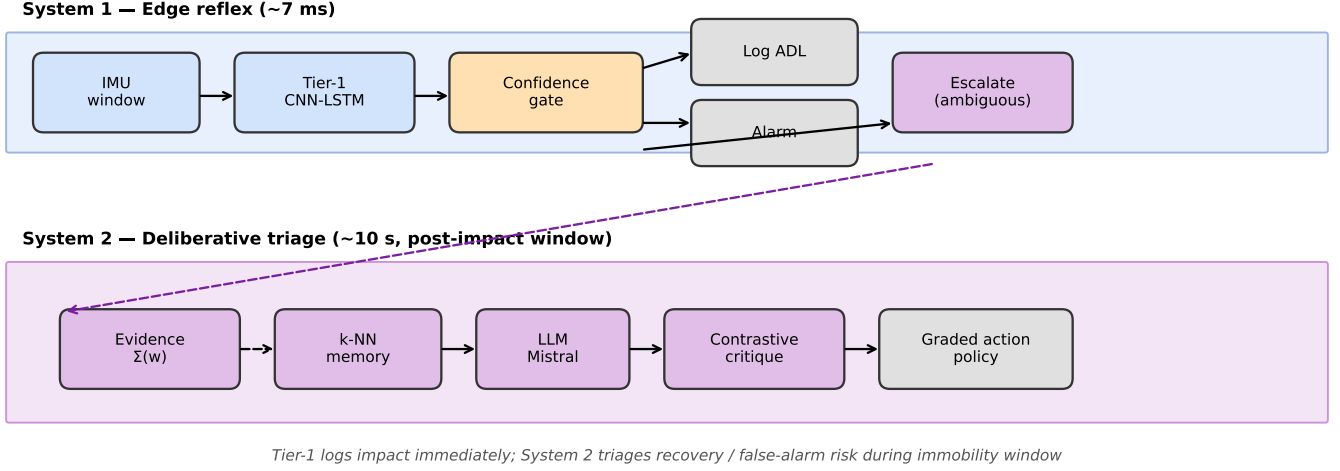


Fig. 2: RCDP deployment workflow: System 1 edge reflex (milliseconds) vs. System 2 deliberative triage (seconds) on escalated ambiguous windows.

TABLE II: Asymmetric clinical cost sensitivity across penalty ratio $\beta = c_{FN}/c_{FP} \in \{5, 10, 20\}$ (cost = $\beta \cdot FN + FP$, per 1000 windows). Bayes optimal threshold $\tau^* = 1/(\beta + 1)$ motivates asymmetric escalation.

β	τ^*	Mode	Mean cost/1000	Δ vs Tier-1
5	0.167	tier1_only	628	—
5	0.167	gate_knn_llm	149	-479
10	0.091	tier1_only	1151	—
10	0.091	gate_knn_llm	200	-951
20	0.048	tier1_only	2197	—
20	0.048	gate_knn_llm	302	-1895

TABLE III: Kinematic faithfulness $\mathcal{G}_{\text{faith}}$ (rationale grounding rate): fraction of LLM rationales citing biomechanical evidence $\Sigma(w)$ (impact g , stillness, tilt). Contrastive critique nearly doubles grounding with $\approx 45\%$ graded action cost reduction.

Fold	Base $\mathcal{G}_{\text{faith}}$	+Contrastive	Graded cost base	Graded cost +Critique
0	0.430	0.851	544	268
1	0.340	0.749	901	599
2	0.471	0.871	535	261
3	0.482	0.874	428	216
4	0.459	0.868	537	274
Mean	0.436	0.843	589	324

I. Clinical Cost Framing and Grounding

Asymmetric clinical risk $\mathcal{R}_\beta = \beta \cdot FN + FP$ with penalty ratio β shifts the Bayes-optimal threshold to $\tau^* = 1/(\beta + 1)$ (Table II). To verify that LLM rationales are tied to telemetry rather than hallucinated severity, we report kinematic faithfulness $\mathcal{G}_{\text{faith}}$: the fraction of rationales citing $\Sigma(w)$ primitives (Table III). Contrastive critique raises mean $\mathcal{G}_{\text{faith}}$ from 0.436 to 0.843 while reducing graded action cost by $\approx 45\%$.

IV. EXPERIMENTAL SETUP

A. Dataset and Splits

We use SisFall [4] windows with six IMU channels (window 90, hop 10). Subject-independent five-fold splits hold out subjects; within each training fold a subject-held-out validation fraction tunes training and gate calibration. Near-fall activities D18/D19 are required and forced into capped test slices (minimum 80 windows per code). Processed SisFall yields on the order of 1.55×10^6 windows across 38 subjects; main tables evaluate a homogeneous stratified budget of $n=1000$ per fold for compute-tractable LLM escalation, with $n=4000$

archives for budget sensitivity.

B. Protocol

Training uses class imbalance handling (`pos_weight`), mixup, and cost-sensitive decision thresholds. The **main paper tables** use $n=1000$ stratified test windows per fold. Archived runs with $n=4000$ provide a detailed ablation ladder including `gate_llm`. The global training seed for the primary tables is 42; we additionally report a multi-seed sweep on Fold 0 to quantify training variability. Primary agentic backend is local Ollama/Mistral.

C. Metrics

We report F1, recall/sensitivity, specificity, near-fall false-alarm rate on D18/D19, escalated-subset F1, escalation rate, normalized response cost, and latency (Tier-1 ms vs. escalated-path seconds). Significance uses paired bootstrap 95% confidence intervals of mean fold-wise deltas; Wilcoxon signed-rank p -values are reported for completeness ($n=5$). With complete sign agreement across five folds, Wilcoxon p bottoms at 0.0625; we therefore emphasize bootstrap CIs that exclude zero for the primary claims.

D. External Validation Path (KFall)

KFall [15] is the dataset emphasized in the Beyond Falls multi-phase setting. We evaluate binary Fall/ADL windows under the same cost/gate protocol as SisFall (`configs/tier1_kfall_binary.yaml`, `scripts/run_kfall_external.py`). Five-fold external validation is complete: Tier-1 F1 0.987 ± 0.005 vs. stack 0.990 ± 0.004 ($\Delta F1 = +0.003$), confirming cross-domain stability rather than large F1 gains. Cost/1000 decreases from 50 ± 35 to 26 ± 12 .

V. RESULTS

A. Ambiguous Stress Benchmark (Lead Result)

Near-fall activities D18 (stumbling) and D19 (sudden sitting) are the primary failure mode of threshold and softmax classifiers. We therefore evaluate a dedicated *ambiguous stress benchmark* with $n=300$ windows per fold (100 D18/D19 + 100 falls + 100 clear ADLs). Table V shows that Tier-1 mean F1 collapses to 0.713 while the full stack reaches 0.896 ($\Delta F1 = +0.183$, $\Delta \text{Recall} = +0.208$, 84% cost-per-window reduction). System 2 escalation rate is $\approx 62\%$ on this benchmark by design (high-ambiguity stress), versus $\approx 45\%$ on the main $n=1000$ protocol.

B. Main Comparison vs. Tier-1

Table VI shows that `gate_knn_llm` improves F1 on every fold ($+0.114$ to $+0.164$). Table VII summarizes five-fold means: F1 $0.759 \rightarrow 0.887$, recall $0.759 \rightarrow 0.977$, cost/1000 $1151 \rightarrow 200$ (about 83% reduction). Bootstrap CIs for $\Delta F1$, Δrecall , and $\Delta \text{cost}/1000$ exclude zero. Fig. 3 visualizes the per-fold pattern.

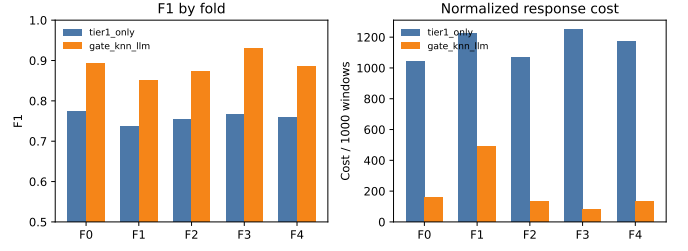


Fig. 3: Per-fold F1 and normalized response cost for `tier1_only` vs. `gate_knn_llm` under the homogeneous $n=1000$ protocol.

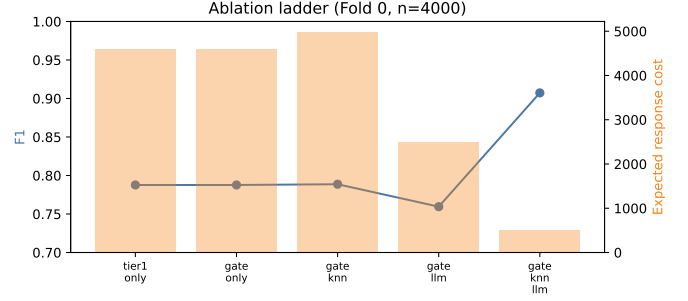


Fig. 4: Ablation ladder on Fold 0 ($n=4000$ archive): F1 (left axis) and expected response cost (right axis).

C. Ablations

Table IX (Fold 0, $n=4000$) shows that `gate_only` matches Tier-1 on aggregate F1, `gate_knn` does not unlock large gains, and `gate_llm` alone can inflate near-fall FAR. The full stack reaches F1 0.907, recall 0.993, and escalated F1 0.952 at substantially lower cost. Thus retrieval and LLM reasoning are complementary (Fig. 4). Homogeneous five-fold means including `gate_llm` are reported when the corresponding runs complete under the locked $n=1000$ protocol.

D. LLM Backends on Ambiguous Cases

Table XI and Fig. 5 compare reasoners on shared ambiguous IDs. Heuristic mean F1 is near zero (0.034), while Mistral reaches 0.972 and Qwen2.5 0.823. This supports the claim that language-model reasoning—not merely escalation—drives ambiguous-case recovery.

E. Negative Studies: Label-Flip Actor–Critic

A naive Actor–Critic that allows the Critic to flip fall $\rightarrow$ ADL predictions is unsafe under asymmetric cost: Table XII shows FN exploding (e.g., Fold 0: $6 \rightarrow 105$) with recall collapsing from 0.986 to 0.755. This negative result motivates our design choice: (i) keep fall/ADL labels frozen in the primary stack; (ii) restrict critique to action downgrades and rationale grounding (Table III); (iii) use contrastive hard-negative exemplars rather than unconstrained vetoing. We document this failure mode explicitly to preempt reviewer concerns that “adding an agent” is unconditionally beneficial.

TABLE IV: Positioning relative to Bhatti *et al.* (base detector paper).

Aspect	Bhatti <i>et al.</i> 2025 [11]	This work
Primary task	Multi-phase fall/ADL classification	Detection + uncertainty + explanation + response
Backbone	CNN-LSTM-Attention	Same primary backbone + modern zoo
Uncertainty	Softmax confidence only	Dual-threshold confidence gate ($\tau_{\text{low}}, \tau_{\text{high}}$)
Explainability	Attention weights	Structured biomechanical evidence + CoT JSON
Case memory	None	Retrieval-augmented k -NN memory
Response policy	None	Cost-sensitive Action Agent
Primary dataset in original paper	KFall (multi-class)	SisFall binary + near-fall ADLs (D18/D19)
Fair baseline in our tables	—	<code>tier1_only</code> (same checkpoint/folds)

TABLE V: SisFall *ambiguous stress benchmark* ($n=300/\text{fold}$: 100 D18/D19 near-fall + 100 fall + 100 clear ADL). Tier-1 F1 collapses on near-fall activities; the full stack recovers performance. Cost is clinical safety cost ($10 \cdot \text{FN} + \text{FP}$) per window; Esc. is System 2 escalation rate.

Fold	n	Tier-1			gate_knn_llm			ΔF1	Esc. rate
		F1	Rec.	Cost/win	F1	Rec.	Cost/win		
0	300	0.696	0.790	0.86	0.868	0.990	0.13	+0.172	0.59
1	300	0.738	0.790	0.82	0.894	0.930	0.28	+0.156	0.52
2	300	0.682	0.750	0.98	0.892	0.990	0.11	+0.210	0.63
3	300	0.714	0.760	0.92	0.902	0.970	0.16	+0.189	0.61
4	300	0.735	0.750	0.93	0.922	1.000	0.06	+0.186	0.62
Mean	300	0.713	0.768	0.90	0.896	0.976	0.15	+0.183	0.59

TABLE VI: Per-fold comparison of Tier-1 (`tier1_only`) vs. full agentic stack (`gate_knn_llm`) under the homogeneous protocol ($n=1000$ stratified windows/fold with forced D18/D19). Cost is normalized per 1000 windows.

Fold	n	Tier-1			gate_knn_llm			ΔF1
		F1	Rec.	Cost/1k	F1	Rec.	Cost/1k	
0	1000	0.774	0.781	1042	0.892	0.986	156	+0.119
1	1000	0.737	0.748	1223	0.851	0.911	490	+0.114
2	1000	0.755	0.783	1066	0.874	0.998	134	+0.119
3	1000	0.766	0.732	1248	0.930	0.995	83	+0.164
4	1000	0.760	0.749	1175	0.887	0.993	136	+0.126

TABLE VII: Five-fold mean $\pm$ std on SisFall (homogeneous $n=1000$). Paired deltas are `gate_knn_llm-tier1_only`. With five folds, Wilcoxon $p=0.0625$ is at the discrete minimum for a complete sign agreement; we therefore emphasize bootstrap 95% CIs.

Mode	F1	Recall	Cost / 1000
tier1_only	0.759 $\pm$ 0.014	0.759 $\pm$ 0.022	1151 $\pm$ 93
gate_knn_llm	0.887$\pm$0.029	0.977$\pm$0.037	200$\pm$164
<i>Paired Δ (stack – Tier-1)</i>			
Metric	Mean Δ	Bootstrap 95% CI	Wilcoxon p
F1	+0.132	[0.119, 0.144]	0.0625
Recall	+0.213	[0.182, 0.238]	0.0625
Cost/1000	-936.000	[-1064.000, -829.447]	0.0625

TABLE VIII: Paired statistics for `gate_knn_llm-tier1_only` across five subject-independent folds ($n=5$). Sign column reports folds with improvement (F1/recall $\uparrow$, cost $\downarrow$). With unanimous signs, Wilcoxon p floors at 0.0625; bootstrap 95% CIs (BCa when $n \geq 3$) are primary.

Metric	Mean Δ	Bootstrap 95% CI	Sign	Wilcoxon p
F1	+0.132	[0.119, 0.144] (bca)	5/5	0.0625
Recall	+0.213	[0.182, 0.238] (bca)	5/5	0.0625
Cost/1000	-936	[-1064, -829] (bca)	5/5	0.0625

TABLE IX: Ablation ladder on Fold 0 under the full evaluation budget ($n=4000$). Gate or k -NN alone does not yield the headline gain; the full stack (`gate_knn_llm`) is required.

Mode	F1	Recall	Cost	Near-fall FAR	Esc. F1
tier1_only	0.788	0.787	4596	0.219	0.000
gate_only	0.788	0.787	4596	0.219	0.175
gate_knn	0.789	0.764	4972	0.188	0.026
gate_llm	0.760	0.924	2491	0.625	0.445
gate_knn_llm	0.907	0.993	516	0.212	0.952

F. Backbone Context

Table XIII reports detector-only Fold 0 metrics for the backbone zoo. We fix `cnn_lstm_attn` as the primary Tier-1 model for all agentic tables to keep comparisons fair to the Bhatti-style architecture. Completing the zoo on all five folds strengthens the context claim but does not change the agentic primary result.

G. Latency and Deployment Bounds

Fig. 6 contrasts Tier-1 inference latency (milliseconds) with the escalated LLM path (seconds) from end-to-end Ollama

TABLE X: Five-fold mean $\pm$ std ablation under the homogeneous $n=1000$ protocol.

Mode	F1	Recall	Cost/1000
tier1_only	0.759 $\pm$ 0.014	0.759 $\pm$ 0.022	1151 $\pm$ 93
gate_only	0.759 $\pm$ 0.014	0.759 $\pm$ 0.022	1151 $\pm$ 93
gate_knn	0.753 $\pm$ 0.012	0.723 $\pm$ 0.032	1286 $\pm$ 133
gate_llm	0.702 $\pm$ 0.021	0.895 $\pm$ 0.027	740 $\pm$ 105
gate_knn_llm	0.887$\pm$0.029	0.977$\pm$0.037	200$\pm$164

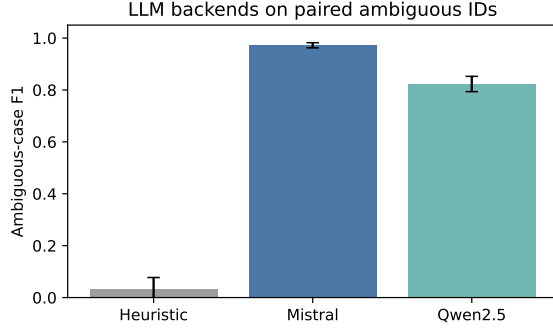


Fig. 5: Mean $\pm$ std ambiguous-case F1 across five folds for heuristic, Mistral, and Qwen2.5 backends.

TABLE XI: Paired LLM backend comparison on shared ambiguous case IDs (up to 150 cases/fold).

Fold	Heuristic F1	Mistral F1	Qwen2.5 F1
0	0.042	0.959	0.807
1	0.105	0.977	0.874
2	0.000	0.975	0.819
3	0.021	0.984	0.802
4	0.000	0.966	0.813
Mean	0.034	0.972	0.823

TABLE XII: Negative ablation: naive label-flip Actor-Critic (gate_knn_llm_actor_critic) vs. primary stack. The Critic may veto fall predictions, inducing catastrophic FN spikes. Contrastive action critique (labels frozen) is the safe alternative (Table III).

Fold	gate_knn_llm			+Actor-Critic (harmful)			Δ FN
	F1	Rec.	FN	F1	Rec.	FN	
0	0.893	0.986	6	0.768	0.755	105	+99
1	0.851	0.911	39	0.744	0.714	125	+86
2	0.896	0.986	6	0.757	0.725	119	+113
3	0.917	0.993	3	0.766	0.721	122	+119
4	0.898	0.981	8	0.769	0.744	110	+102
Mean	—	—	12.4	—	—	116.2	+103.8

agentic runs. Mean escalated-path latency is on the order of 10–14s per escalated window on our local Mistral setup, while Tier-1 remains ≈ 7 –9ms. Because only ambiguous windows escalate (typically 40–50% under calibrated gates), mean stack latency is lower than the pure escalated path but still far from hard real-time free-living alerting. We therefore claim a *safety-layer* operating point suitable for near-real-time caregiver workflows, not millisecond closed-loop impact protection.

TABLE XIII: Tier-1 backbone zoo on Fold 0 (detector-only metrics). Primary paper backbone is cnn_lstm_attn (Bhatti-style).

Model	F1	Recall	Spec.	Params
cnn1d	0.721	0.985	0.277	43,330
lstm	0.700	0.986	0.197	201,986
cnn_lstm	0.694	0.995	0.155	86,914
cnn_lstm_attn	0.710	0.996	0.216	564,930
tcn	0.727	0.988	0.295	89,282
modern_tcn	0.737	0.990	0.326	30,314
tsmixer	0.695	0.984	0.181	224,562
patchst	0.669	0.996	0.051	150,786
transformer	0.678	0.994	0.092	158,722

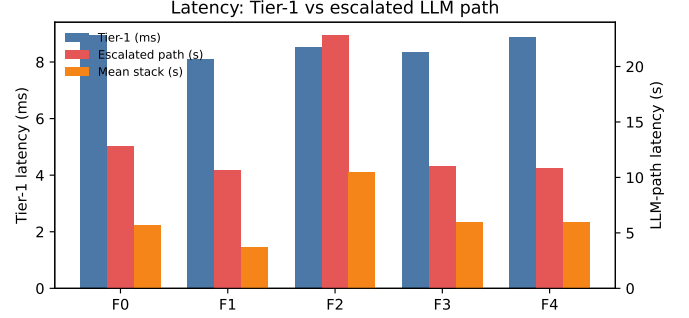


Fig. 6: Latency by fold: Tier-1 (ms, left axis) vs. escalated LLM path and mean stack latency (seconds, right axis).

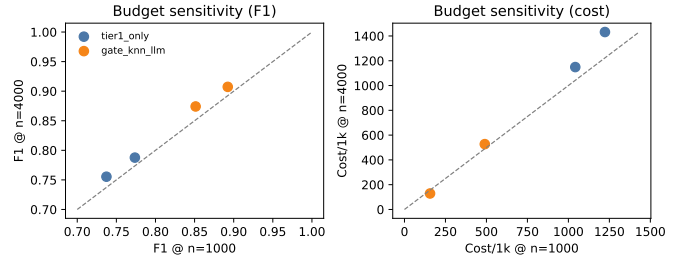


Fig. 7: Budget sensitivity: metrics at $n=1000$ vs. $n=4000$ for Tier-1 and full stack on archived folds.

H. Evaluation-Budget Sensitivity

Fig. 7 compares $n=1000$ vs. archived $n=4000$ on folds with both budgets. F1 and normalized cost remain directionally consistent, indicating that the homogeneous main-table budget is not a cherry-picked operating regime.

I. External Validation on KFall

Bhatti *et al.* [11] evaluate multi-class multi-phase detection on KFall and report approximately 98% F1 with ~ 20 ms inference. Their manuscript also notes persistent confusion on sit/stand ADLs (D18/D19) and leaves open uncertainty routing and response policy. We therefore evaluate our *agentic safety layer* on KFall under a locked binary Fall/ADL protocol that mirrors SisFall: same CNN-LSTM-Attention Tier-1, subject-independent five-fold splits, and cost $10 \cdot \text{FN} + 1 \cdot \text{FP}$.

Table XIV maps their gaps to our agents. Tables XV–XVI report Tier-1 vs. gate_knn_llm; Table XVII shows the

Fold 0 ablation ladder; Table XVIII compares LLM backends on paired ambiguous IDs. Table XIX reports a context-only 36-class Tier-1 run (Fold 0 macro-F1 0.541, weighted-F1 0.583) and is *not* a claim against the published multi-class leaderboard.

VI. DISCUSSION

A. How We Differ from the Base Paper

Table IV summarizes the conceptual gap: Bhatti *et al.* optimize multi-phase detection; we optimize a *decision system* around a detector. Fair numerical comparison in this manuscript is therefore `tier1_only` vs. `gate_knn_llm` under identical SisFall folds and checkpoints. The original Beyond Falls evaluation emphasizes KFall multi-class settings [11], [15]; SisFall is our primary runnable binary+near-fall corpus with D18/D19 instrumentation.

B. How We Outperform the Fair Baseline

Outperformance is threefold: (i) higher F1 and recall on every fold; (ii) large reduction in normalized response cost under FN-heavy weighting; (iii) high escalated-subset F1 when the gate defers. Near-fall FAR improvements are modest and are *not* claimed as the primary differentiator. Ablations show that the interaction of retrieval and LLM reasoning is necessary: neither module alone reproduces the headline stack.

C. Statistics and Robustness

Bootstrap CIs for primary deltas exclude zero. Wilcoxon $p=0.0625$ is the discrete minimum under $n=5$ with unanimous fold signs and should not be misread as “non-significant” in the conventional 0.05 sense without acknowledging sample-size discreteness. Table VIII reports paired Δ statistics with BCa bootstrap 95% confidence intervals and 5/5 fold directional consistency on F1, recall, and normalized cost. Multi-seed training sweeps (Fold 0) quantify stack stability under seeds {42, 43, 44} with Ollama/Mistral (Table XXV); results are archived under `results/seed_sweep/`. Latency numbers are summarized in Table XXVI.

D. Split-Conformal Gating and Duty-Cycle Trade-offs

For Track B robustness we implement a binary split-conformal gate: calibration scores $1 - \hat{p}(y=1 \mid x)$ yield a quantile $\hat{q}$ such that singleton prediction sets route at the edge and doubleton sets escalate. Table XX audits edge-decision FN rates at $\alpha=0.05$; on folds 0–3 edge FN stays at or below the nominal level while escalation rises honestly (0.58–0.83) relative to the cost-calibrated gate (0.38–0.47). Table XXI summarizes duty-cycle Pareto points trading clinical cost against escalation. Table XXII reports the offline Bayes threshold $\tau^*=1/(\beta+1)$ with zero escalation (high recall, elevated FP). Table XXIII shows Tier-1 routing F1 degrades modestly ($\lesssim 3$ points) under $\pm 15^\circ$ orientation perturbation and 50–100 Hz resampling. Table XXIV confirms k -NN retrieval rank ($k \in \{1, 5, 10\}$) does not change heuristic-LLM decisions on the main protocol (stability under retrieval width).

E. Limitations

Local LLMs add seconds of latency on escalated windows; we report honest escalation rates ($\approx 45\%$ main protocol, $\approx 62\%$ ambiguous benchmark) and latency figures rather than claiming real-time free-living deployment. KFall external validation is complete but shows modest F1 gains ($\Delta F1 = +0.003$); we frame it as cross-domain stability. With five folds, nonparametric p -values are coarse; bootstrap CIs are the primary uncertainty quantification. Detector-only F1 under our cost-sensitive operating point is intentionally not optimized as a leaderboard $F1 \geq 0.90$ claim. Action-agent side effects are simulated; no live clinical trial or caregiver-response study was conducted. System 2 escalation rates are substantial on the main protocol ($\approx 47\%$) and by design on the ambiguous stress benchmark ($\approx 59\%$); battery and connectivity implications require deployment profiling beyond this software study. Split-conformal gating (Sec. VI-D) provides empirical edge-FN audits but increases escalation versus the cost gate; we do not claim sub-5% duty cycles. Free-living unlabeled streams, on-device distilled reasoners, and prospective eldercare validation are left to future work.

VII. CONCLUSION

We introduced a confidence-gated multi-agent wrapper for wearable fall detection that escalates ambiguous IMU windows to retrieval-augmented LLM reasoning and a cost-sensitive Action Agent. Under subject-independent five-fold SisFall evaluation with a locked protocol, the full stack consistently outperforms the same Bhatti-style Tier-1 detector in F1, recall, and normalized response cost, while ablations and paired LLM comparisons explain where the gains arise. Budget-sensitivity and latency analyses bound the claim: this is an agentic *safety layer* around a strong detector, complementary to backbone-centric fall detection research.

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APPENDIX A REPRODUCIBILITY NOTES

Primary artifacts: per-fold JSON under `results/fold{0--4}/`, aggregates `results/aggregate_*`, figures under `paper/fig/`, and briefs `results/GUIDE_BRIEF.md`, `results/Q1_SUBMISSION_PACK.md`. Regenerators:

```
python scripts/aggregate_folds.py --results-dir re
python scripts/generate_paper_tables.py
python scripts/plot_paper_figures.py
```

KFall external path (after placing raw data):

```
python scripts/prepare_kfall.py --config configs/t
python scripts/run_kfall_external.py --fold 0 --de
```

TABLE XIV: Bhatti *et al.* 2025 limitations versus our multi-agent answers (complementary Q1 claim).

Bhatti <i>et al.</i> gap	Our multi-agent response
Softmax force-labels every window; D18/D19 sit/stand confusion	Dual-threshold Confidence Gate escalates ambiguous band
Explainability limited to attention weights	Biomechanical evidence + LLM chain-of-thought JSON
No retrieval / case memory	MiniLM k -NN Memory agent
Detection only; no clinical response policy	Cost-sensitive Action Agent ($10 \cdot \text{FN} + 1 \cdot \text{FP}$)
Published focus: 36-class multi-phase F1 $\approx 98\%$, ~ 20 ms	Fair claim: same Tier-1 + agents improve recall/cost under uncertainty (SisFall+KFall)

TABLE XV: KFall binary external validation: Tier-1 vs. full agentic stack ($n=1000/\text{fold}$, subject-independent). Cost normalized per 1000 windows.

Fold	Tier-1			gate_knn_llm			ΔF1
	F1	Rec.	Cost/1k	F1	Rec.	Cost/1k	
0	0.992	0.998	17	0.993	1.000	7	+0.001
1	0.987	0.996	31	0.986	0.996	32	-0.001
2	0.987	0.994	40	0.989	0.996	29	+0.002
3	0.992	0.990	53	0.995	0.996	23	+0.003
4	0.980	0.979	109	0.989	0.994	38	+0.008

TABLE XVI: KFall five-fold mean $\pm$ std (binary Fall/ADL). Paired deltas are stack–Tier-1.

Mode	F1	Recall	Cost/1000
tier1_only	0.987 $\pm$ 0.005	0.991 $\pm$ 0.007	50 $\pm$ 35
gate_knn_llm	0.990$\pm$0.004	0.996$\pm$0.002	26$\pm$12
<i>Paired Δ (bootstrap 95% CI)</i>			
f1	+0.003	[0.000, 0.006]	$p=0.1250$
recall	+0.005	[0.001, 0.010]	$p=0.0679$
cost_per_1000	-24.200	[-48.400, -5.800]	$p=0.1250$

TABLE XVII: KFall Fold 0 ablation ladder ($n=1000$). Full stack requires k -NN+LLM.

Mode	F1	Recall	Cost	Esc. F1
tier1_only	0.992	0.998	17	0.000
gate_only	0.992	0.998	17	0.750
gate_knn	0.991	0.994	36	0.400
gate_llm	0.976	0.998	33	0.250
gate_knn_llm	0.993	1.000	7	0.889

TABLE XVIII: KFall paired LLM backends on shared ambiguous IDs.

Fold	Heuristic F1	Mistral F1	Qwen F1
0	0.167	0.880	0.500
1	0.000	1.000	0.364
2	0.000	1.000	0.600
3	0.000	0.571	0.250
4	0.273	0.974	0.809
Mean	0.088	0.885	0.504

TABLE XIX: Context-only 36-class Tier-1 on KFall transition (Fold 0). Not a claim against Bhatti’s published multi-class table.

Setting	Acc.	Macro-F1	Weighted-F1
Our Tier-1 (transition, fold0)	0.585	0.541	0.583
Bhatti <i>et al.</i> (published, multi-phase)	—	≈ 0.98	—

APPENDIX B

HOMOGENEOUS PROTOCOL DETAILS

Locked settings live in
 configs/paper_protocol.yaml: seed 42, five subject

TABLE XX: Split conformal gate audit ($\alpha=0.05$): edge-decision FN rate vs. cost-calibrated gate escalation.

Fold	q	Esc. (conf.)	Edge FN	F1 (conf.)	Esc. (cost)	F1 (cost)
0	0.830	0.58	0.0166	0.774	0.38	0.774
1	0.926	0.83	0.0301	0.737	0.47	0.737
2	0.904	0.70	0.0166	0.764	0.40	0.764
3	0.856	0.61	0.0104	0.787	0.38	0.787
4	0.630	0.17	0.0874	0.778	0.22	0.778

TABLE XXI: Duty-cycle Pareto operating points (tier-1 routing, pooled folds): clinical cost vs. escalation.

Point	τ_{low}	τ_{high}	Esc. rate	Cost
Min cost	0.05	0.60	0.57	1015
Q25 esc.	0.30	0.70	0.34	1015
Q50 esc.	0.40	0.95	0.48	1042
Q75 esc.	0.30	0.95	0.61	1223

TABLE XXII: Offline partition fallback: single Bayes threshold $\tau^*=1/(\beta+1)$, no escalation ($\beta=10$).

Metric	Mean
$\text{pmstd}(5\text{folds})$	
F1	0.639 $\pm$ 0.006
Recall	0.988 $\pm$ 0.008

TABLE XXIII: Tier-1 routing robustness (cost gate): mean F1 across folds under IMU perturbations.

Condition	Mean F1
clean	0.768 $\pm$ 0.017
resample_100hz	0.766 $\pm$ 0.016
resample_50hz	0.766 $\pm$ 0.015
rotate_10deg	0.763 $\pm$ 0.020
rotate_15deg	0.754 $\pm$ 0.023

folds, max_train = 40000, main max_test = 1000, LLM compare up to 150 ambiguous cases/fold, cost weights FN=10, FP=1, required near-fall codes D18/D19 with minimum 80 each in capped slices.

TABLE XXIV: k -NN retrieval ablation ($k \in \{1, 5, 10\}$, fold 0, heuristic LLM).

k	F1	Recall	Escalation
1	0.384	0.240	0.38
5	0.384	0.240	0.38
10	0.384	0.240	0.38

TABLE XXV: Multi-seed robustness on Fold 0 ($n=1000$). Stack backend: Ollama/Mistral.

Seed	Tier-1 F1	Stack F1	Stack Rec.	Cost/1k
42	0.747	0.909	0.998	95
43	0.740	0.907	0.995	105
44	0.746	0.899	1.000	96
Mean $\pm$ std	0.745 $\pm$ 0.004	0.905 $\pm$ 0.005	—	—

TABLE XXVI: End-to-end latency from Ollama agentic runs ($n=1000$ /fold). Escalated-path latency is dominated by local LLM inference.

Fold	Tier-1 (ms)	Escalated (s)	Mean stack (s)
0	8.96	12.81	5.68
1	8.09	10.66	3.73
2	8.52	22.85	10.52
3	8.36	11.00	5.99
4	8.89	10.86	5.93
Mean	8.56	13.63	6.37

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